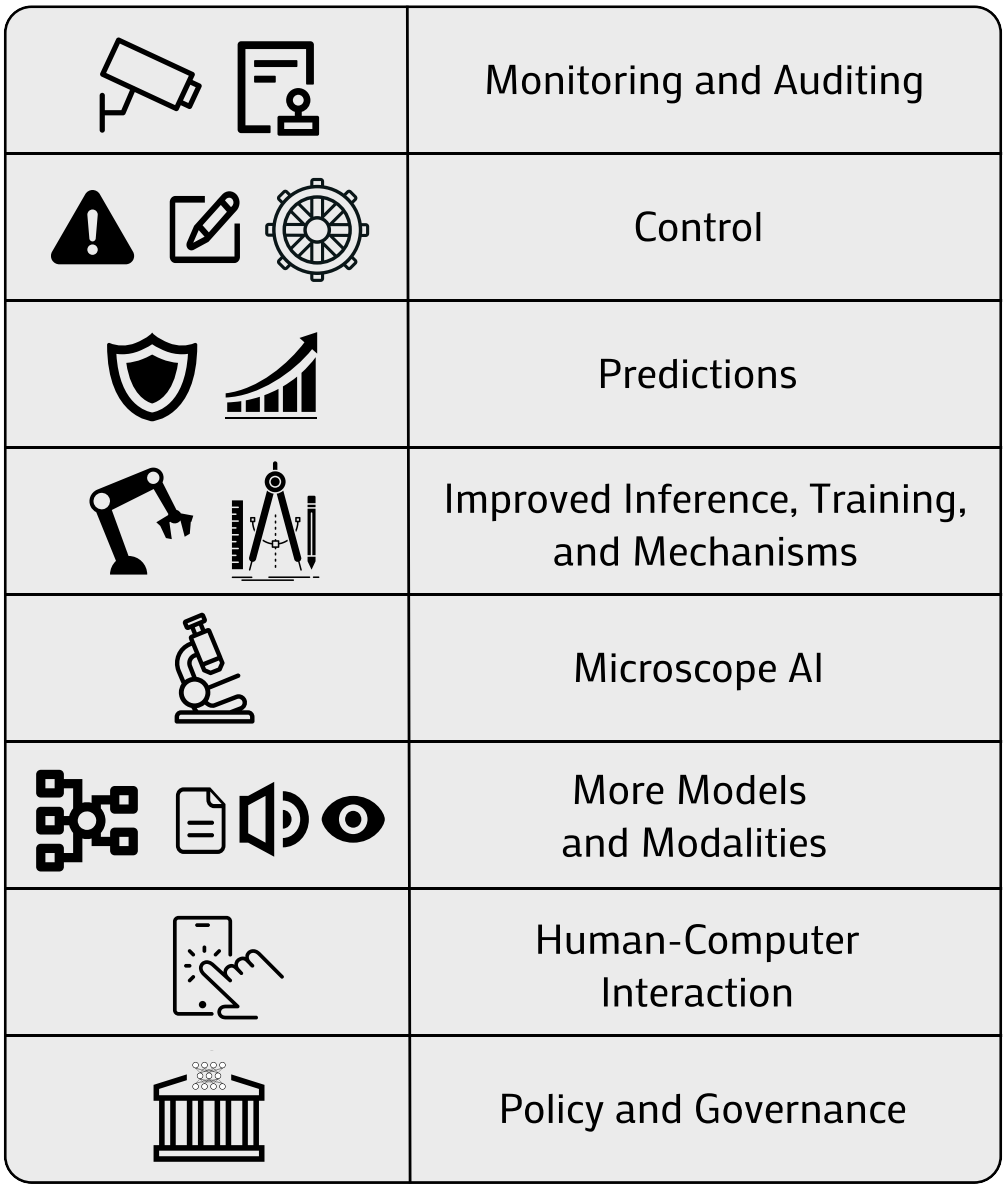

Figure 6: A summary of problem areas for applications of mechanistic interpretability.

(Laine et al., 2024); or giving dishonest responses tailored to match the user's beliefs ("sycophancy"; Sharma et al. (2024b)). Interpretability techniques could be used to uncover the mechanisms underlying these potentially harmful behaviors and thus help to detect and characterize them. This becomes increasingly important as the capabilities of models increases, especially when using training methods that incentivize models to feign particular properties for the purpose of passing evaluation.

Using interpretability methods to identify internal signs of concern (also known as "white-box" evaluations (Casper et al., 2024) or "understanding-based" evaluations (Hubinger, 2023)) is therefore an important problem. White-box evaluation methods could serve as tools to detect potential biases that arise when models learn to use spurious correlations (Gandelsman et al., 2024a; Casper et al., 2022a; Abid et al., 2022). However, human judgment might be required to determine which features are 'supposed' to be relevant to the task (Marks et al., 2024; Kim et al., 2018a; Goyal et al., 2022).

Despite current shortcomings in decomposition and description, white-box evaluations are likely feasible today. Even shallow, correlation-based descriptions could signal potentially concerning cognition. For example, developing new methods that reliably distinguish between features that merely recognize deceptive behavior vs. mechanisms that cause deceptive behavior may be challenging. However, a correlation-based method that flags both can facilitate catching the latter. To be useful, it may not even be necessary to decompose or describe the entire network; having descriptions for components that are used on concerning subdistributions of model behavior might suffice. For instance, imperfect interpretability methods may help evaluators develop hypotheses about how models will behave, thus guiding further inquiry. Meanwhile, recent work proposes incorporating SDL (Paragraph 2.1.2c) into safety cases for advanced AI systems. By monitoring internal representations, it could aid in detecting potential sabotage or deceptive behavior before deployment (Grosse, 2024). While such approaches show promise, they have difficulty in validating whether learned features capture all concerning patterns of reasoning reliably.

Evaluations for unsafe cognition may be a particularly important use case as it plays well to the comparative advantages of mechanistic interpretability relative to the other areas of machine learning. The majority of other areas of machine learning already focus on controlling or steering the behavior of AI systems to alter input-output behavior. It is therefore unclear that this is to mechanistic interpretability's comparative advantage. On the other hand, interpretability is perhaps the only research area that attempts to understand the mechanisms of model cognition. This implies that it might be particularly fruitful for interpretability researchers to tackle problems that become easier to solve through improving such understanding: auditing for unsafe cognition, debugging unexpected behavior, and monitoring systems in deployment.

3.2.1a Enabling real time monitoring of AI systems for potentially unsafe cognition

Beyond white-box evaluations, interpretability has further applications in monitoring. For instance, internals could be used to passively monitor the system during deployment, much like content moderation systems currently in use today. Alternatively, internals could be used to flag when a model takes an action for abnormal reasons even in absence of satisfactory descriptions (Section 2.1.3), known as "mechanistic anomaly detection" (Christiano, 2022; Johnston et al., 2024), which may be a sign of suspicious behavior. Mechanistic anomaly detection primarily requires progress in decomposition methods (Section 2.1.2) as it is necessary to be confident about what constitutes an individual mechanism within the network. Current SDL methods identify active latents, but not active mechanisms, which are implemented by network parameters. It may not be necessary to have deep descriptions of the function of individual mechanisms as long as anomalies can be detected.

3.2.1b Improving our ability to red-team AI systems and elicit unsafe outputs

Beyond white-box evaluations, leveraging interpretability could improve our ability to conduct adversarial attacks, red-team, or jailbreak AI systems (Casper et al., 2024). This process is beneficial as it exhibits failure modes models may display in the wild, when facing adversarial pressure from wide deployment or malicious actors, thereby enabling developers to effectively preempt and address them. Furthermore, it may form a significant element of safety cases (Clymer et al., 2024; Balesni et al., 2024a; Grosse, 2024; Goemans et al., 2024) for AI systems, by providing assurances of form: We tried hard to red-team the system, yet failed to exhibit concerning behavior despite having *more* affordances than users may have. One reasonable assumption is that developers may have white-box access to models, while users may not. Although many existing red-teaming methods (Perez et al., 2022; Zou et al., 2023b) already require gradient access, we could additionally leverage interpretability insights to accelerate red-teaming. For instance, Arditi et al. (2024) discovered a universal "refusal direction" in chat-finetuned language models, which is causally important for models engaging in the behavior of refusing harmful requests. Lin et al. (2024) used this to red-team models by optimizing for inputs that minimize the projection of the residual stream onto this direction during the forward pass. This approach may be more efficient than optimizing over the whole model, as previous methods did (Zou et al., 2023b). Mechanistic interpretability techniques also promise to improve our ability to attribute model outputs to their corresponding inputs paragraph 2.1.3a. This could enhance human red-teamers' ability to find key input features responsible for bad behavior in models, thus speeding

up iteration cycles. Though such approaches are possible today with only feature-based understanding, they might improve with more crisp mechanism-based understanding.

3.2.2 Using mechanistic interpretability for better control of AI system behavior

Ensuring the safe deployment of AI first requires effective control over their behavior. Currently, the techniques used for this purpose are mostly unrelated to interpretability (e.g. Christiano et al. (2017); Rafailov et al. (2023); inter alia), but sometimes inspired by it (Rimsky et al. (2024), Zou et al. (2024), Kirch et al. (2024); inter alia). Mechanistic interpretability could assist in interpreting (Lee et al., 2025) and improving (Conmy & Nanda, 2024) these control methods, or in developing new ones. In this section, we outline interpretability-inspired control methods, and envision future possibilities with further progress in interpretability methods.

One new control method derived from mechanistic interpretability insights is **activation steering** (a.k.a. activation addition). A fixed activation vector, hypothesized to linearly represent a model concept, is added to an intermediate activation of a model at inference time (Li et al., 2024a; Turner et al., 2024; Zou et al., 2023a; Rimsky et al., 2024). Turner et al. (2024) introduced activation steering, directly inspired by the Linear Representation Hypothesis (discussed in Paragraph 2.1.2c). This technique results from the hypothesis, and its success can be thought of as evidence for the hypothesis. Moderate success can be achieved in steering using basic decomposition and description methods. Mechanistic interpretability decomposition methods enable the steering of models toward a narrower range of behaviors with fewer side effects (Chalnev et al., 2024). Advancements in mechanistic interpretability methods are likely to result in improved steering capabilities, such as activating entire mechanisms instead of individual features.

Machine unlearning was originally defined as the problem of scrubbing the influence of particular data points on a trained machine learning model (Cao & Yang, 2015). In the context of modern generative models, machine unlearning is more broadly defined as removing particular undesirable knowledge or capabilities ('unlearning targets') from models, while preserving model performance on tasks involving non-targets (Liu et al., 2024a). Targets for unlearning that are of particular interest include sensitive private or copyrighted data, model biases (Liu et al., 2024b), and hazardous knowledge that could be misused by malicious actors (Li et al., 2024b); for instance, information regarding the creation of bioweapons. A better understanding of how knowledge or capabilities are implemented within model internals can help in the development of new techniques for machine unlearning (Belrose et al., 2023b; Zou et al., 2024; Guo et al., 2024; Pochinkov & Schoots, 2024; Ashuach et al., 2024), as well as to better evaluate unlearning efficacy through white-box, non-behavioral, techniques (Lynch et al., 2024; Deeb & Roger, 2024; Hong et al., 2024). Thus far, mechanistic interpretability methods that modify intermediate activations (but not weights) for unlearning have yet to yield competitive results (Farrell et al., 2024).

Unlearning falls under the broader aim of **model (knowledge) editing**, which seeks to make precise modifications to a machine learning model that incorporates specific knowledge with desirable generalization properties, while minimizing the impact on other knowledge (Wang et al., 2024c). By attempting to carve neural networks at their joints, mechanistic interpretability could improve our ability to make interventions on knowledge with few side effects. Meng et al. (2022a) make initial progress toward interpretability-based model editing with their ROME technique. However, Thibodeau (2022) and Hase et al. (2023) highlight flaws in the technique, indicating that mechanistic interpretability has not yet found appropriate model components to intervene on (Section 2.1.2). With better comprehension of neural networks, we should anticipate more surgical model editing techniques in the future.

Editing any given capability or piece of knowledge presents a greater challenge than deleting them. Meaningful progress in unlearning and editing methods may depend on improved network decomposition methods, as it would require isolating the individual mechanisms that correspond to specific knowledge or capabilities. Progress in mechanistic interpretability may elucidate the structure of knowledge and capabilities in AI models, leading to a better understanding of what kinds of model edits possibilities are realistic in future. Knowledge and capabilities could, in fact, be part of large mechanisms that overlap with each other, making it challenging to isolate them into discrete components. For mechanistic interpretability to effectively guide